

Final Project

Renzo Paredes

2025-04-26

Libraries

```
library(readxl)
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    3.5.1      v tibble     3.2.1
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.0.4
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

Loading Data

```
Budget_vs_Spending_Data <- read_excel("C:/Users/renzo/OneDrive/Escritorio/DATA 211-01 Data Science and V
View(Budget_vs_Spending_Data)

head(Budget_vs_Spending_Data)
```

```
## # A tibble: 6 x 3
##   Date                Daily_Budget Daily_Spending
##   <dtm>                <dbl>         <dbl>
## 1 2025-04-06 00:00:00      42           64.4
## 2 2025-04-07 00:00:00      39           61.2
## 3 2025-04-08 00:00:00      43           35.4
## 4 2025-04-09 00:00:00      47           44.8
## 5 2025-04-10 00:00:00      38           28.5
## 6 2025-04-11 00:00:00      38           33.5
```

Data Preparation

Prepare the dataset by converting the Date column to the correct type and reshaping the data for visualizations.

```
Budget_vs_Spending_Data$Date <- as.Date(Budget_vs_Spending_Data$Date, format = "%m/%d/%Y")

long_data <- pivot_longer(Budget_vs_Spending_Data,
                           cols = c(Daily_Budget, Daily_Spending),
                           names_to = "Type",
                           values_to = "Amount")

head(long_data)
```

```
## # A tibble: 6 x 3
##   Date      Type      Amount
##   <date>    <chr>    <dbl>
## 1 2025-04-06 Daily_Budget    42
## 2 2025-04-06 Daily_Spending  64.4
## 3 2025-04-07 Daily_Budget    39
## 4 2025-04-07 Daily_Spending  61.2
## 5 2025-04-08 Daily_Budget    43
## 6 2025-04-08 Daily_Spending  35.4
```

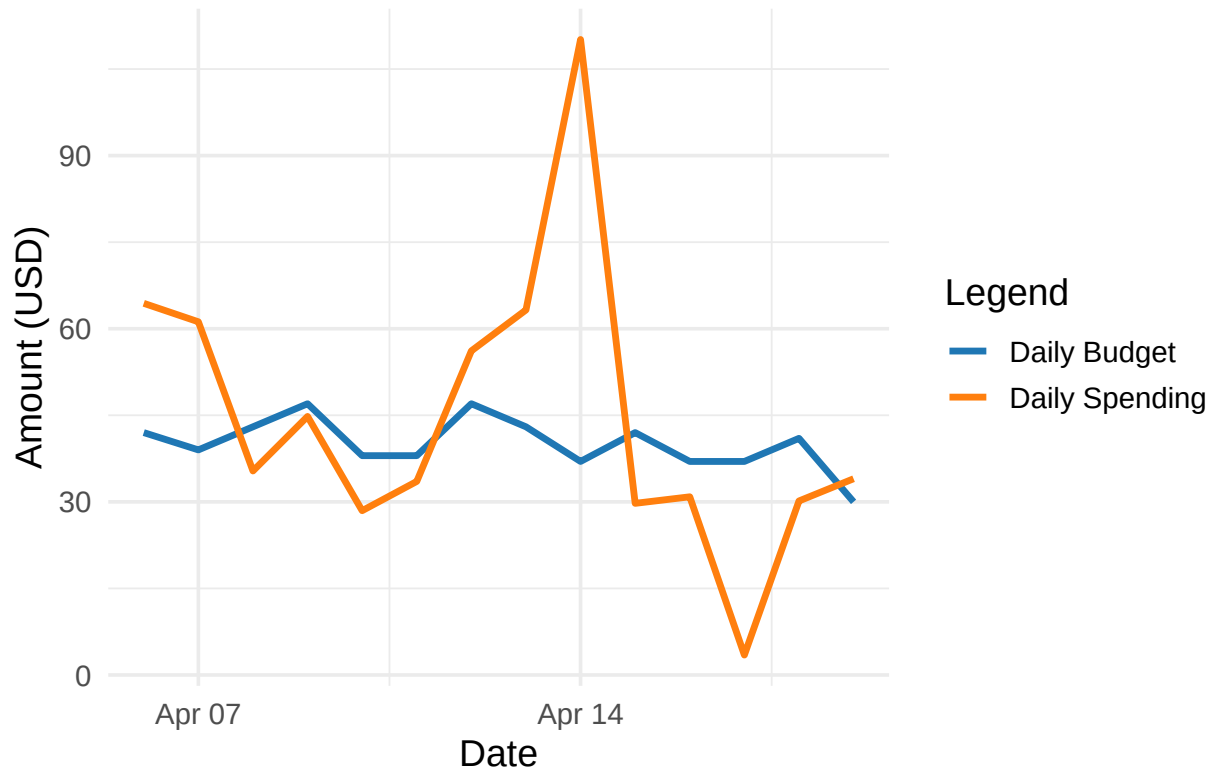
Line Plot - Daily Budget vs. Daily Spending

Visualize how your daily budget and spending change over time.

```
ggplot(Budget_vs_Spending_Data, aes(x = Date)) +
  geom_line(aes(y = Daily_Budget, color = "Daily Budget"), size = 1.2) +
  geom_line(aes(y = Daily_Spending, color = "Daily Spending"), size = 1.2) +
  scale_color_manual(values = c("Daily Budget" = "#1f77b4", "Daily Spending" = "#ff7f0e")) +
  labs(title = "Daily Budget vs. Daily Spending",
       x = "Date", y = "Amount (USD)", color = "Legend") +
  theme_minimal(base_size = 14)
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

Daily Budget vs. Daily Spending



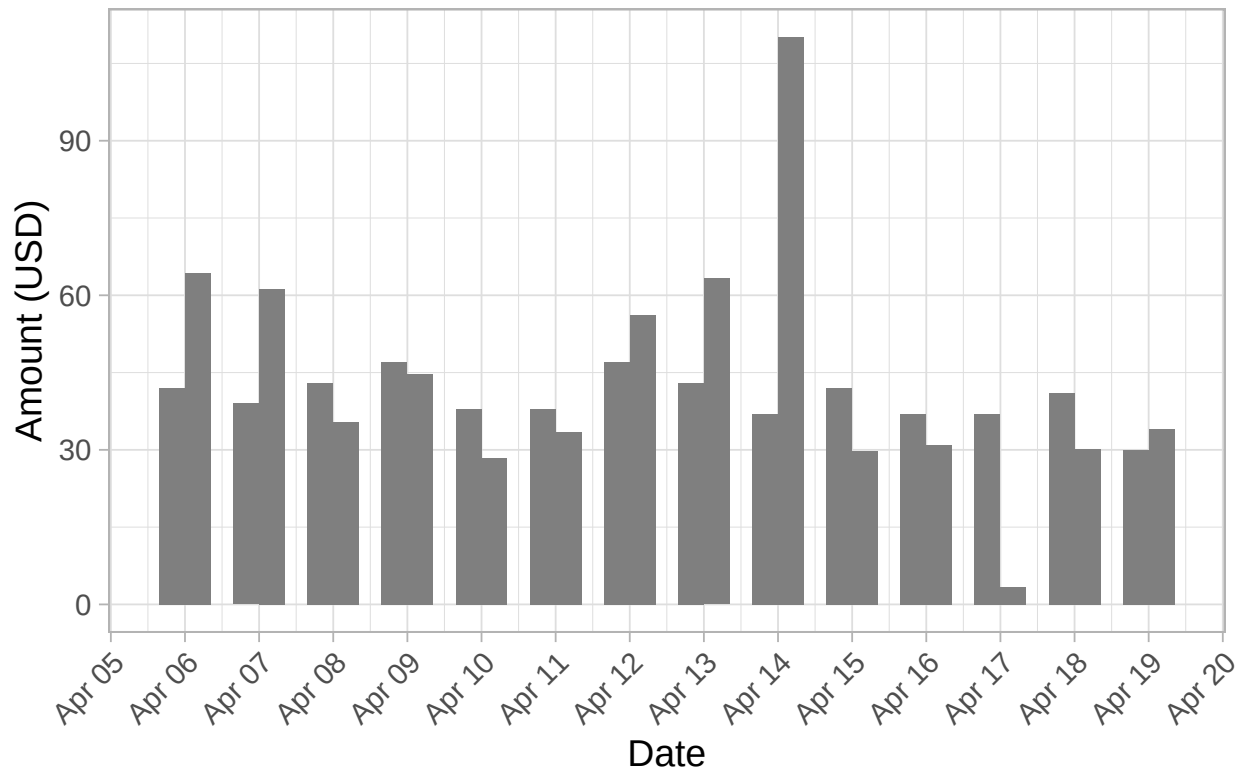
Bar Plot - Daily Comparison

Compare daily budget and daily spending side-by-side for each day.

```
ggplot(long_data, aes(x = Date, y = Amount, fill = Type)) +
  geom_col(position = "dodge", width = 0.7) +
  scale_fill_manual(values = c("Daily Budget" = "#1f77b4", "Daily Spending" = "#ff7f0e")) +
  labs(title = "Daily Budget vs. Daily Spending (Side-by-Side)",
       x = "Date", y = "Amount (USD)", fill = "Type") +
  theme_light(base_size = 14) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_x_date(date_breaks = "1 day", date_labels = "%b %d")
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's fill values.
## No shared levels found between 'names(values)' of the manual scale and the
## data's fill values.
```

Daily Budget vs. Daily Spending (Side-by-Side)



Boxplot - Distribution and Outliers

Understand the distribution and detect potential outliers in your spending and budgeting.

```
ggplot(long_data, aes(x = Type, y = Amount, fill = Type)) +  
  geom_boxplot(alpha = 0.7, outlier.color = "red", outlier.size = 3) +  
  scale_fill_manual(values = c("Daily Budget" = "#1f77b4", "Daily Spending" = "#ff7f0e")) +  
  labs(title = "Distribution of Budget and Spending",  
        x = "", y = "Amount (USD)") +  
  theme_minimal(base_size = 14)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the  
## data's fill values.  
## No shared levels found between 'names(values)' of the manual scale and the  
## data's fill values.
```

Distribution of Budget and Spending



Statistical Analysis: Paired t-Test

We want to test whether the mean Daily Budget is significantly greater than the mean Daily Spending. Since the data points are related (budget and spending for the same day), we use a paired t-test.

$$H_0 : \mu_{\text{Daily Budget}} \leq \mu_{\text{Daily Spending}}$$

$$H_a : \mu_{\text{Daily Budget}} > \mu_{\text{Daily Spending}}$$

Where:

- $\mu_{\text{Daily Budget}}$ = Mean of the Daily Budget
- $\mu_{\text{Daily Spending}}$ = Mean of the Daily Spending

This is a **one-tailed paired t-test**.

```
t_test_result <- t.test(Budget_vs_Spending_Data$Daily_Budget,  
                        Budget_vs_Spending_Data$Daily_Spending,  
                        paired = TRUE,  
                        alternative = "greater")
```

```
t_test_result
```

```
##
```

```
## Paired t-test
##
## data: Budget_vs_Spending_Data$Daily_Budget and Budget_vs_Spending_Data$Daily_Spending
## t = -0.68809, df = 13, p-value = 0.7483
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
## -16.46459      Inf
## sample estimates:
## mean difference
## -4.607143
```

Conclusion

Since the p -value = 0.7483 is greater than the significance level $\alpha = 0.05$, we fail to reject the null hypothesis H_0 . At the 5% significance level, we do not have enough evidence to conclude that the average Daily Budget (μ_{budget}) is significantly greater than the average Daily Spending (μ_{spending}).